Retention, accumulation, and the finite ledger

Chapter II / Source-bound development draft
Sean Brady / Originator and conceptual director
OpenAI ChatGPT and Codex / Research collaborators and co-authors

Abstract. A retained account can outlast its present address. This chapter develops that proposition through original Write, interacting-clock, spin, and outgoing-field results. It separates predictive state, timed history and energetic complement, then supplies general shared-source spectral relations, a finite-record discriminator, an exact memory-closure theorem, and a driven reference instrument joining signed Writes to an unchanged actual clock. Original-source realizability gives a further record payment. Physical preparation, source incidence and calibration remain explicit receiving assignments.

1. The physical proposition

SAM begins with matter displacing substrate and the displaced account being retained. The aquarium thought experiment makes this relation visible: an object appearing in a filled enclosure changes the surrounding arrangement. Its physical content is the requirement to account for the consequence and its later reception. A fluid equation is an additional model choice. [1]

The recent work lets us ask this more precisely. What must a Write leave behind so that its consequence remains available to a later interaction? A present reading, a persistent source state, an ordered history, and an outgoing field can retain different parts of that account.

This chapter uses an operational distinction: two histories remain distinguishable when a specified permitted continuation gives different receiver outputs. This defines what a receiving description must preserve. It does not assign a measured energy or a spatial density to information alone.

The finite ledger organizes the available types. Its physical payload is developed through the source, evolution, reference and receiver that consume it.

2. Three meanings of a retained record

A record can be a current address, a state sufficient for future prediction, or an accumulated history statistic. These meanings already separate in native execution. One retained J4 source has eight immediate outputs but sixteen predictive states. Requiring the next Write output refines eight classes to fourteen; another step gives sixteen. Every distinct pair has a separating word of length at most two. [3]

The same source supplies two 324-Write histories with the same endpoint, eight-bin occupancy, and scalar logarithmic summaries. Their first-half receiver values differ: approximately 0.1440964305 and 0.4063809938 in the declared X direction. Reordering closed blocks changes when a response occurs while preserving its whole-history occupancy. [3]

The eight immediate outputs in this assay and the eight partition addresses introduced later have separate source definitions. Their equal cardinalities supply no transition map.

Chapter I contributes another exact distinction. Its four equal-occupation source eigenstates have zero D and C currents while carrying different energies. A current record therefore leaves an energetic complement. A complete energy receiver must retain that complement even when the current is zero. [2]

The task is consequently to specify the record needed by the chosen future receiver, then carry its physical account through an actual evolution.

3. Predictive equivalence and retention

For a fixed source, event alphabet and receiver, let O(h;u) denote the output obtained by extending history h with an allowed future word u. Define

$$h \sim h' \iff O(h;u) = O(h';u) \text{ for all } u.$$

This relation is preserved by appending the same event: each subsequent test is already a permitted continuation. A quotient state can therefore preserve every future output when its classes and successor transitions obey this relation. The completed sixteen-state assay gives an explicit finite realization. [3]

A clocked record requires more. Suppose an admitted event from q has positive actual phase increment f(q,e). Preserving its entire future phase trace requires preservation of the present output, every f(q,e), and the successor class. Repeated partition refinement gives the coarsest such quotient in a finite instrument. This is a new receiving theorem; its phase increments are supplied source operands.

A two-state construction illustrates the distinction. Both states display one address and alternate under the sole event. Their successive phase increments are 1,2 and 2,1. Both have phase 3 after two events, yet their first recorded phases differ. An untimed quotient has one class; its phase-compatible quotient has two.

4. A clock-weighted history monitor

Chapter I supplies actual positive rates on a stated preparation and time interval. With continuous lifted phase theta and rate nu,

$$d\theta = \nu(\tau)d\tau, \qquad 0 < \ell \le \nu \le u.$$

For a piecewise retained label j, define its elapsed exposure H, phase exposure P, and first phase moment M:

$$H_j = \int 1_j \, d\tau, \quad P_j = \int 1_j \, d\theta,$$

$$M_j = \int \theta 1_j \, d\theta.$$

A constant-label segment with actual phase endpoints alpha,beta updates P by beta-alpha and M by (beta squared minus alpha squared)/2. Prescribed phase gates use the intersection of their phase intervals with this segment. No constant within-segment rate is needed. The achieved rate bounds give

$$\ell H_j \le P_j \le u H_j.$$

For a receiver affine in phase, these counters determine its full integral exactly. Event counts, elapsed exposure and phase exposure remain separate measures. The counter family is finite for the specified receivers; its numbers can grow with the history.

The completed two-past-coordinate covariance recipient also survives a matched-node change to actual phase coordinates. Its original physical lags remain attached. Substituting raw nonlinear phase differences for those lags changes the receiver. [4]

5. Original Write action and occupation

Chapter I retains lifted coordinates

$$\gamma = \varphi_a + \varphi_b - 2\varphi_p,$$

$$\delta = \varphi_a - \varphi_b, \qquad \chi = \varphi_p,$$

$$\omega_W = \hbar(d\gamma + 3d\chi).$$

Its full pair source has swap S, occupation projector N, and even operator K_e:

$$N = (I + S)/2, \qquad H_p = 2N + K_e.$$

K_e commutes with N and both currents. It is distinct from the positive interacting field form also called K in Chapter I. [2]

For a fixed normalized source preparation psi and n equal to its N expectation, declare the receiving lift

$$\Psi_W = e^{-3i\chi} e^{-i\gamma N}\psi.$$

Its canonical one-form is exactly

$$i\hbar\langle\Psi_W|d\Psi_W\rangle = \hbar(n\,d\gamma + 3d\chi).$$

The original one-form is recovered on the occupied class N psi=psi. The original delta coordinate and ordered word remain in the retained record; this two-coordinate receiving lift supplies no independent delta generator. It also retains the distinction between the successful current-observable map and the literal endpoint Hilbert action. [5]

6. The full even-energy payment

The full energy lift at the same current angle is

$$\Psi_H = e^{-i\gamma H_p/2}\psi.$$

Because N and K_e commute, its relative operator is

$$\Psi_W = e^{-i(3\chi I - \gamma K_e/2)}\Psi_H.$$

On a sharp K_e sector the relative factor is a scalar phase. Across coherent sectors it carries their distinct phases. For kbar equal to the mean even energy and V its variance, the squared generator residual is

$$\|R\|^2 = \frac{\dot\gamma^2}{4}V + \left(\frac{\dot\gamma}{2}\bar k - 3\dot\chi\right)^2.$$

This separates a mean-phase payment from an operator complement. Scalar canonical one-form matching is a weaker, separately defined question:

$$3d\chi = (n - 1 + \bar k/2)d\gamma.$$

For the actual placement vector, n=1, kbar=573047, and V=14384109824. Thus the occupied canonical lift is exact while the full source-energy lift retains a nonzero complement.

Original legal role tangents fix the gamma/chi ratio. Their direct identification with unshifted H_p/2 requires an additional reference or channel payment. These explicit conditions locate the remaining physical incidence; selecting chi afterward would change the original path.

7. The actual clock preparation

In a driven lane N=(I+sigma_x)/2 and Z=sigma_z+i sigma_y. Positivity of every reduced source state gives

$$|\langle Z\rangle|^2 \le 4n(1-n).$$

A nonzero clock amplitude therefore requires 0<n<1. The occupied class in Section 5 has zero clock amplitude and remains a separately labelled example.

The completed hard preparation instead coherently combines exact charge-zero and charge-one branches. Its charge expectation is 1/2 and variance 1/4. On their charge orbit,

$$z(\gamma) = Me^{-i\gamma}, \qquad a = \frac{\hbar}{2}d\gamma.$$

The transition phase reads a charge difference; its canonical action reads the prepared charge expectation. For actual hard branch energies E0 and E1, their gap Omega gives z=M exp(-i Omega tau), while the canonical energy payment includes E0+Omega/2. [2]

The full interacting clock retains both field polarizations and its actual joint correlator. In a general changing phase frame its connection is

$$a = \hbar(n\,d\gamma + 3d\chi + a_\eta).$$

The internal joint-state Berry form a_eta carries changing preparation and entanglement. For finite-energy continuum evolution its canonical energy identity is understood through the form-domain overlap derivative.

8. A worked actual phase history

Use the completed lane01 hard-coherent/soft-vacuum preparation and its actual phase lift theta(0)=0. Its outward rate bounds, per model time, are

$$3.490102929541\times10^{-6} \le \nu,$$

$$\nu \le 3.490102934505\times10^{-6}.$$

The native certificate retains sharper exact rational endpoints. Throughout T0=573049 pi, theta is increasing. The first three quarter-phase records occur uniquely at theta=k pi/2, k=1,2,3, with

$$\frac{k\pi}{2u} \le \tau_k \le \frac{k\pi}{2\ell}.$$

Existence follows from 573049 ell>3/2. The fourth record is outside this certified window because 573049 u<2. The bare cycle therefore retains three completed quarter crossings and a partial fourth; it is not a completed actual phase turn.

Each record carries k, its wrapped quarter address, lifted theta, and elapsed-time enclosure. For the conserved total joint energy E_tot, its dynamical energy action is hbar E_tot tau_k. The complete variational action also carries the Berry form and both counterterms.

These are actual clock-threshold records, separately typed from original executed Writes and the eight partition addresses. Their source-to-Write identification is the physical instrument still to specify. The same construction applies to all six achieved clock certificates.

9. A pulse whose boundary returns

The retained regular-interior/outgoing-exterior wave action supplies a concrete account of transport. Choose one normalized monopole, positive R,kappa,c, and a prescribed clamped source pulse J=q on [0,tau_p), zero otherwise. Let H=2R/c. [6]

Its exact impulse kernel and boundary coordinate are

$$g_0(t) = \frac{c}{R} 1_{(0,\,H)}(t),$$

$$a(t) = \frac{qc}{\kappa R}\,\mathrm{length}([0,\tau_p] \cap [t-H,\,t]).$$

The overlap rises, may plateau, and returns to zero at tau_p+H. The bare ON coupling work is zero because a(0)=0. Release supplies the exact complete cycle work

$$W = \frac{q^2}{\kappa R}\min\left(\frac{c\tau_p}{R}, 2\right) > 0$$

for nonzero q. An inherited stored counterterm contributes at ON and release and telescopes over the full OFF-to-OFF cycle. Both endpoint payments remain in the account.

This is a prescribed-control mean-energy example under one common stationary Gaussian reference. It retains its control cost and does not replace the autonomous microscopic source evolution.

10. Where the retained pulse resides

The boundary OFF energy is

$$E_b(t) = \frac{\kappa a(t)^2}{4R}.$$

The full field energy while ON is q a(t)/R; after release it is W. At every time,

$$E_{\mathrm{field}} = E_b + E_{\mathrm{radial+kinetic}},$$

$$E_{\mathrm{radial+kinetic}} \geq 0.$$

After tau_p+H the boundary mean is zero. Deterministic forcing preserves the stationary Gaussian covariance, so its whole reduced boundary marginal then matches the unforced reference. The outgoing bulk mean field still carries W. Independent spatial integration of that pulse gives the same energy.

For the declared normalization R=kappa=c=q=1 and tau_p=1, the boundary coordinate rises to one, falls to zero at t=3, and W=1. Its peak boundary energy is 1/4. At late times all unit energy is in the bulk complement. These numbers illustrate the exact formula; they assign no physical source scale.

Persistent source memory is a second form of retention. In the inherited source law the written f_H contribution remains in J and continues to sustain a displaced field. A finite response horizon therefore need not erase either an outgoing account or a persistent source record. [6]

11. The spin complement and memory

The latest spin calculation completes J21, F11 and the retained T5 for all 54 preparations. It supplies a quantitative complement rather than an unspecified missing account. [7]

Let v=A w_d and let Pi project onto a specified retained source span. The source inner product gives

$$v = \Pi v + r, \qquad \langle \Pi v, r \rangle = 0,$$

$$\|v\|^2 = \|\Pi v\|^2 + \|r\|^2.$$

The completed source-action excess uses the six-function span (d,e,u_d,u_e,w_d,w_e). The next lower-response direction instead uses the five-function span ending at w_d. Their residuals retain their separate projections. Physical energy conversion retains the actual energy operator B:

$$\langle v, Bv \rangle = \langle \Pi v, B\Pi v \rangle + \langle r, Br \rangle$$

$$+2\mathrm{Re}\, \langle \Pi v, Br \rangle.$$

The mixed term remains unless the receiving projection is B-orthogonal.

A finite source also gives an exact dynamical continuation. Write u'=-Au, p=Pi u, q=(I-Pi)u. Solving the q equation and substituting it into p gives a retained evolution with a memory integral and an initial-hidden-state term. Eliminating a coordinate changes the evolution law unless those contributions vanish. The derivation is retained in the technical packet.

12. The cost of a specified record

Select one common positive source energy B and a linear receiver r=L^T y. The positive displacement cost is y^T B y. Put G=L^T B^{-1}L. For an attainable record,

$$\min_{L^T y = r} y^T B y = r^T G^+ r.$$

The minimizing vector is B^{-1}L G^+r. Every other feasible vector adds a nonnegative B-norm. On constrained microscopic source states this ambient minimum remains a lower bound; equality requires that vector to be physically realizable.

The inherited literal eighteen-coordinate energy and a declared diagnostic receiver give an exact executable instance. This receiver is separately labelled from the original complete receiver family. [8]

A positive displacement cost and a signed energy change have different references. Relative to y0,

$$\Delta E = 2y_0^T B\, \delta y + \delta y^T B\, \delta y.$$

The native example has positive displacement cost 2006049/573049 and signed energy change -1017407/573049. The mixed reference payment explains the signs.

For an ensemble, the complete mean energy additionally retains Tr(B covariance). A source-dependent projection minimum, a common reference energy, and the universal floor therefore keep their distinct types.

13. The structural finite ledger

The structural account receives distinction arity h=2 and spatial support d=3. Their fixed-binary closure d^(d-1)=2^d+1 has the unique integer solution d=3 for d at least two. At d=2 it fails; for d at least four, d^(d-1) is greater than 2^d+1.

The radix and bounded product kernel are

$$\mathcal{R} = h^2 d = 12,$$

$$\mathcal{B} = \{h^a d^b : 0 \le a \le 3,\ 0 \le b \le 2\}.$$

Its full grid is

a	b=0	b=1	b=2
0	1	3	9
1	2	6	18
2	4	12	36
3	8	24	72

Retaining values at most twelve gives P={1,2,3,4,6,8,9,12}. The smallest admitted overflow is Theta=18. Sixteen requires exponent a=4 and lies outside this kernel. The source caps and route grammar are part of the construction. Ordinary multiplication alone is not a complete physical Write law. [9]

The completed-WRITE route also supplies R=M_w D_route=4 times 3. Route depth and spatial support have distinct meanings, and its boundary assumptions remain attached. [9]

14. Carrier shares and the floor

The square account is N_c=R squared=144. Its carrier and retained complement are

$$\Theta = 18 = \frac{N_c}{8}, \qquad N_{\mathrm{ret}} = 126 = \frac{7N_c}{8}.$$

The binary face-state route assigns one unresolved carrier state among 2^d states. This explains its structural one-eighth share. A source energy fraction additionally requires the energy and receiving operators developed above. The closed ledger gives 162-18=144=126+18. [9]

The floor combines separately typed compact cycle, arity and support:

$$A_0 = \frac{1}{(2\pi)hd} = \frac{1}{\pi\mathcal{R}} = \frac{1}{12\pi}.$$

The equality between 2h and h squared uses h=2. The source also registers the spherical projection A_H/(4 pi d), with unit local closure A_H=1. Its angular allocation and geometry remain explicit. [9]

One conditional energy receiver is

$$A_{\mathrm{rec}} = A_0 + \frac{E_{\mathrm{disp}}}{E_*}, \qquad E_* > 0.$$

Here E_disp is a selected positive common-reference displacement cost. This equation supplies a candidate physical interpretation of lift once its source incidence and scale are selected. It supplies neither a radial law nor a new numerical determination of A0. Preparation-dependent spin norms remain separately normalized.

15. The common source and its two-level lanes

Keep the signed-response K_r in Q=T+g_H K_r distinct from the commuting pair-energy operator K_e. Let alpha=epsilon_phys/hbar be the positive physical frequency scale. One complete unequal-occupation lane has the same Hamiltonian, full field, counterterms and evolution as a two-level source with those same operands. The common source supplies its further relations across lanes and occupation sectors. [2,11]

Removing only Q_L Q_R from the complete pair Hamiltonian gives the additive control S+Q_L+Q_R. The removed term is scalar within each driven lane, so all six inherited current clocks are identical under this control. A complementary receiver reads its intersector energy instead.

The protected diagonal states have D=C=0, including their source counterterms. Their exact frequencies are

$$\omega_{ba} = \alpha(q_b - q_a)(q_b + q_a + 2).$$

The same receiver in the additive control gives 2 alpha(q_b-q_a). The proposed three-state coherent preparation produces two adjacent frequency terms, with the same amplitude and one common scale. The source preparation and its interference receiver remain explicit physical inputs.

16. A uniform comparison and shared calibration

Order the four roots q0<q1<q2<q3. In placement order (1,0,2,3), Q(g_H) is an irreducible real Jacobi matrix for every g_H at least zero. Its roots are simple; diagonal dominance gives

$$68 + 544g_H \le q_a \le 100 + 2112g_H.$$

Let R_pair=omega21/omega10 and let R_add be the corresponding additive ratio. Then

$$\frac{R_{\mathrm{pair}}}{R_{\mathrm{add}}} = \frac{q_2 + q_1 + 2}{q_1 + q_0 + 2} > 1 + \frac{1}{132}.$$

The bound follows from interlacing, q2-q0 at least 8+32g_H. It compares the two sources at the same g_H. A specified compact calibration interval gives a sharper margin. It does not assign every parameter the same sampling cadence or distinct adjacent frequencies.

The six bare signed couplings supply

$$\Xi = \sum_{a<b} \frac{g_{ab}^2}{\Delta^2}$$

$$= 384 + 5120g_H + 1377280g_H^2.$$

This function is strictly increasing for g_H at least zero. Within the retained source law it identifies g_H from a dimensionless readout; one absolute scale then calibrates epsilon. The protected coherence ratio can be retained as a separate prediction rather than independently adjusted.

17. Receiving the hidden source offset

Current-lane couplings determine root differences. A complementary coherence ratio can also recover their common offset. Put

$$a = q_1 - q_0, \quad b = q_2 - q_1,$$

$$\rho = R_{\mathrm{pair}}/(b/a).$$

Then

$$q_0 = \frac{1}{2}\left[\frac{a+b}{\rho - 1} - a - 2\right].$$

Three oriented couplings along a spanning tree, their common bare gap, and this ratio consequently reconstruct the four source eigenvalues in the declared model. Placement eigenvectors remain a separate receiving account. This inverse is an algebraic identity; its denominator shows the increasing sensitivity as rho approaches one. [11]

Source access also has structure. Either endpoint placement in the Jacobi path reaches all four modes for every nonnegative g_H. Interior placement0 instead misses one mode exactly at

$$g_H = \frac{1}{164}.$$

Native exact closure returns dimension three there, while endpoint placement1 returns four. Thus source observability depends on preparation even though the Hamiltonian remains nondegenerate. This new exception was followed from Duck's first-round cyclicity breadcrumb.

18. Eight samples and a finite error allowance

For an autonomous closed two-level source, a fixed preparation and linear receiver give equally spaced samples

$$y_n = c + a\cos(n\vartheta) + b\sin(n\vartheta).$$

First differences remove c. From eight samples form d_n=y_(n+1)-y_n and the four-by-four Hankel matrix H_ij=d_(i+j). Every such two-level record has rank at most two. The protected source generally supplies four distinct sampled phase factors. A nonzero three-by-three minor already distinguishes the two records. [11]

The certified diagnostic uses independently declared nominal source/scale calibration to fix cadence. It includes Xi(1) plus or minus one and relative alpha uncertainty plus or minus one part per million. For the selected minor,

$$-0.106707 < \det H_{013} < -0.085302.$$

The conservative determinant perturbation allowance is below 0.051216 when each sampled expectation has absolute error at most 0.0001, including two competing error records. The target and stated two-level comparison remain separated. Exact rational native rank/inverse checks independently receive the enclosed sample construction.

These are model error allowances for the specified receiver, not achieved measurement precision. Preparation leakage, sensing and physical calibration must supply their own contribution to that allowance.

19. When retained evolution closes

Let A be the finite self-adjoint positive source generator, Pi an orthogonal retained projection, and Q=I-Pi. Write C=Q A Pi and B=Pi A Pi restricted to V=range(Pi). All compressed evolution identities below act on V. The exact retained memory kernel is

$$M(t) = C^{*} e^{-QAQt} C.$$

Exact autonomous retained evolution for every initial preparation is equivalent to each of: C=0, M(0)=0, and invariance of the retained span under A. The same equivalence holds when all preparations start inside that span. A second-order comparison gives

$$\Pi e^{-At}\Pi - e^{-Bt} = \frac{t^2}{2} C^{*} C + O(t^3).$$

For a retained vector v,

$$\langle v, M(0)v\rangle = \|(I-\Pi)Av\|^2.$$

A projection residual therefore supplies the initial dynamical memory strength. The smallest invariant enlargement is the span of V,AV,A squared V, and all subsequent powers until exact rank stabilizes. A small residual supplies an approximation estimate; a zero residual supplies exact closure. [11]

For the spin gap-dependent estimates, the ambient space is the invariant exchange-odd sector. Its even stationary mode is outside this positive-gap account.

20. The actual spin account needs two directions

The completed five-function spin span is

$$b_5 = (d, e, u_d, u_e, w_d).$$

Its only potentially nonzero outgoing columns correspond to u_e and w_d. The initial memory Gram on those columns has the form

$$M_{u_e u_e} = \delta, \quad M_{u_e w_d} = s,$$

$$M_{w_d w_d} = R_5, \qquad \delta R_5 - s^2 > 0.$$

Completed original U,W,J11,J21,F11 contractions certify this strict inequality for all 54 preparations. The kernel therefore has rank two. Every exact invariant enlargement containing b5 needs at least seven dimensions. Its first enlargement is

$$\mathrm{span}(b_5, w_e, r_{\mathrm{next}}),$$

$$r_{\mathrm{next}} = A w_d - \Pi_5 A w_d.$$

That seven-dimensional span is the first Krylov enlargement; its full invariance is a further question. Adding w_e alone gives the completed b6 floor span. Adding r_next alone gives another six-dimensional span. Each leaves the other independent direction.

The retained source gives both R5 and the b6 residual norm between 1.420936 and 1.421320. With the correct retained Gram normalization, the memory norm is below 7.524585. Hence the compressed semigroup defect at source time 0.001 is below 0.000003763. All original N, Gibbs preparations and attempt clocks remain attached. [7,11]

21. The nonzero-clock reference account

The inherited amplitude bound constrains a simple occupation-compensating reference. Put r=1-n. If the actual clock amplitude is at least m>0, then

$$\frac{1-\sqrt{1-m^2}}{2} \le r \le \frac{1+\sqrt{1-m^2}}{2}.$$

The explicit reference state

$$|r\rangle = \sqrt{n}\,|0\rangle + \sqrt{1-n}\,e^{-i\gamma}|1\rangle$$

has connection (1-n)d gamma, including when n changes differentiably. Its real amplitude-derivative terms cancel. It restores the occupation coefficient in the original receiving lift while retaining the internal joint-state Berry form. It does not automatically supply that remaining form. [11]

For actual lane01 the simple reference occupation is confined approximately to 0.499975039 through 0.500024961. This is a recipient-specific action bound, not a universal minimum energy for every reference device.

The next construction receives the entire actual joint-state contribution. It uses a balanced reference and its measurable relative phase, without requiring a new strong differentiability assumption on the continuum state.

22. A paid reference preserving the actual clock

Retain the unchanged Chapter I Hamiltonian H and its actual finite-energy trajectory Psi(t). Here t is model time and E is its conserved total energy-form expectation. For a prescribed legal source path define

$$w = \dot{\gamma} + 3\dot{\chi},$$

$$h_R(t) = 2[w(t) - E]P_R.$$

Start the separate reference qubit balanced. Its relative phase is

$$\beta(t) = 2[\gamma(t) - \gamma(0)$$

$$+3\chi(t) - 3\chi(0) - Et].$$

The composite trajectory is Psi(t) tensor (|0>+exp(-i beta)|1>)/sqrt(2). Its source-field marginal is exactly the inherited actual clock. Its total canonical connection is

$$E\,dt + \frac{1}{2}d\beta = d\gamma + 3d\chi.$$

Physical energy is hbar H/t_u, with t_phys=t_u t. E retains even source energy, fields, correlations and both counterterms. The extra reference phase is a relative qubit phase with an explicit Hamiltonian; it is not a discarded scalar rephasing.

This is a driven receiving apparatus. It gives a constructive coexistence of the signed Write connection and nonzero actual clock signal. The original microscopic source actuator and an autonomous controller remain distinct physical assignments. [11]

23. Preserve the original source duration

Duck's source retrieval recovered the original positive role-path clock. With phase progress ell increasing on either signed stroke,

$$v_j = \frac{d\ell}{dt} = \frac{ct_u}{R_H}(1 - A_{\mathrm{path}})\mathcal{D}_j,$$

$$\mathcal{D}_j^{-1} = \frac{12\sqrt{E_B^2 + g_j^2}}{E_B^2 + 36}.$$

The source contact E_B and role density are inherited. The signed ledger rate is w=sigma v_j; reverse strokes retain positive elapsed duration. The reference construction therefore consumes the original source-paced path directly. A wait has w=0 while the actual clock continues. [11]

For M active quarters and 0<=A_path<=A_max<1, a sufficient finite-window condition is

$$\frac{MR_H}{ct_u}\frac{3}{2\sqrt{5}\,(1 - A_{\max})} \le 573049.$$

Waits add sum(t_wait)/pi to this displayed inequality, whose durations have been divided by pi. Neither R_H nor t_u is assigned a new physical value.

A second, explicitly controlled schedule places M events at actual phase thresholds 3k pi/(2M), all inside the completed window. It agrees with the original source timing only when v_j=(M/3)nu(t). Event ordering, original gamma, positive clock phase and their compatibility remain separate recorded quantities.

24. Switch work and the complete action

On the balanced reference,

$$E_R = w - E, \qquad \mathrm{Var}(h_R) = (w - E)^2.$$

Reference drive power is dot w. At a sharp change the signed work is w_after-w_before. Entry, interstroke and release switches telescope when the drive starts and ends off, while every individual payment stays in the history. The source-path actuator has its own energy account. [11]

A negative Write has negative action relative to this common reference and positive elapsed duration. On a bounded finite protocol h_R remains self-adjoint and bounded below. Adding a scalar baseline to make it nonnegative changes both the canonical connection and its energy action. An opposing reference must receive that baseline if the original absolute ledger is retained.

The total variational form still subtracts the Hamiltonian expectation. It vanishes on this exact driven Schroedinger trajectory in its own lift. Equality of its canonical part to the original ledger does not supply an additional nonzero on-shell variational action.

Thus the apparatus makes the price of the relative phase explicit. Its physical implementation can be developed without changing the mathematical classification of the original occupied lift or the achieved source-field clock preparations.

25. A minimal specified history monitor

For phase-polynomial receivers through degree d, retain moments

$$M_{j,k} = \int \theta^k 1_j \, d\theta, \quad 0 \le k \le d.$$

A constant-label segment alpha to beta updates its kth moment by (beta^(k+1)-alpha^(k+1))/(k+1). The same rule handles a declared phase gate after intersecting its interval with the segment. It uses actual phase endpoints, with no constant-rate approximation. [11]

Within additive linear monitoring and linear reconstruction, the minimum independent counter count is the rank of the specified receiver weights on admissible occupation measures. Known endpoint identities may reduce that rank. The receiver family and permitted histories therefore determine the sufficient record size.

Finitely many moments need not answer every later gate. Closed legal blocks can be reordered while retaining an entire specified lower-moment family. The explicit original-source words in the next section make the remaining distinction visible. A finite predictive state, finite prescribed history statistics, and a complete retrospective history retain their different roles.

26. Same endpoint and first moments, different history

Use original closed legal blocks A=W1+ repeated four times and B=W7+ repeated four times. Compare ABBA with BAAB, each sixteen Writes. They return to the same source state and preserve point counts. Under a prescribed equally spaced phase-event monitor their first moments also agree, while second moments differ. [11]

The stronger source-time statement preserves actual positive block durations. Let T_A,T_B be their durations in a fixed repeated source context; retain each block's internal event offsets. Both histories have total duration 2(T_A+T_B), the same point counts and first physical-time atomic moments. On each of three A-only points,

$$\Delta M_2 = 2T_B(T_A + T_B) > 0.$$

Every internal event offset cancels from this difference. It therefore survives unequal source durations. The common point supplies its separately retained signed contribution.

Prouhet closed-block reorderings extend the prescribed phase-event construction beyond every finite polynomial degree. These event-atomic statistics remain distinct from elapsed label exposure. A nonlinear change from physical time to actual phase transports the receiver and its original time arguments; equality of time moments is not silently promoted to equality of phase moments.

27. Physical realizability adds a definite payment

Section 12 gives the ambient minimum for positive B and record r=L^T y. Let y_star be that minimizer and let F_r contain only microscopic source displacements that actually realize r. Orthogonality gives

$$\inf_{y\in F_r} y^T B y = r^T G^+ r$$

$$+\inf_{y\in F_r} \|y - y_\star\|_B^2,$$

$$G = L^T B^{-1} L.$$

The second term is the realizability correction. It is zero exactly when the ambient minimizer can be attained or approached in the permitted source set. An empty set means the selected record is unreachable under those permissions. [11]

For a finite source the deterministic payment can be enumerated. If random mixtures are admitted and only the mean record is fixed, the minimum expected payment is a finite linear program. Mean displacement alone is insufficient: the covariance contribution remains in the expected quadratic energy.

These are operational source costs under one reference and receiving operator. They do not assign the universal structural floor from a choice of energy units.

28. An actual original-source feasible record

Use the inherited eighteen-coordinate literal energy, all signed cross terms, and the actual original finite port-flip source. A separately declared diagnostic receiver selects record (-1,0). [8,11]

Payment divided by kappa_E	Value
Ambient minimum	0.987626515
Actual source minimum	1.894317938
Realizability correction	0.906691423

The exact actual source cost is

$$C_{\mathrm{source}}/\kappa_E = 1085537/573049.$$

Two actual single-port flips, z2 and z8, attain it. The full fifteen-endpoint contact Gram is calculated natively from B and their original displacement columns. It retains every pair cross term; the feasible minimum is not inferred from an independent-coordinate approximation.

This diagnostic is distinct from the original complete receiver family and from the autonomous continuum preparation. It demonstrates how an allowed microscopic source set changes the price of a specified distinction. A physical accumulation readout may consume this cost only with its selected source incidence, reference and energy scale.

29. The operation determines the readout

A uniform A0 remains in the total account while gradients and shared endpoint differences remove it. A raw lapse ratio instead rescales its lifts through the background normalization. A source-normalized local lapse declares a different input. Each readout retains its operation, reference and domain. [1,9]

The new source comparison sharpens the same lesson. All six current clocks leave the commuting pair-product energy unread. Protected coherence receives that energy, and a source-bound coupling family links its prediction to one common calibration.

The memory theorem supplies the dynamical version: a retained address evolves autonomously exactly when its span is invariant. The actual spin residual identifies two independent outgoing directions. History moments and physically feasible record costs then specify what a later receiver retains and pays.

The controlled reference construction carries the original signed Write form together with an unchanged actual clock. Its full energy, reference switches and source timing stay explicit. The next physical construction is the source actuator, preparation/readout incidence and autonomous controller that receive these mathematical assignments.

30. Sources and reproducibility

The continuation followed the requested order: shared-source signature and calibration, retained-memory closure, then the paid Write/clock instrument and its record/energy recipients. The GEN6 Duck Connections cycle used two 180-second exploration windows with complete shared review between them. Original laws, completed runs and the installed runtime remain preserved. [11]

[1] Owner-supplied Chapter II draft and verification ZIP, 15 October 2026.

[2] Final Chapter I and six actual source-field clocks; H002628/H002630.

[3] Native history reduction; H002165.

[4] Observed-history clock/ruler recipient; H002559.

[5] Original Write/tensor and action-unit map; H002623.

[6] Radiative memory and complete work; H002319.

[7] Completed spin source-action T5; H002627.

[8] Common spin energy and receivers; H002423.

[9] Original Chapter II source packet CR001/002/003/005 and local CR113/114/115/218/238.

[10] First Chapter II development packet; H002634.

[11] voli_chapter2_continuation2_20261015: source-bound derivations, Duck cycle receipts, independent checks, reader sources and portable replay. New constructions: Codex. Sean Brady: originator/conceptual director. OpenAI ChatGPT and Codex: research collaborators and co-authors.